

Yilin Wang

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RESEARCH INTERESTS

Seeking **Research Scientist / Applied Scientist / ML Engineer** roles in autonomous driving and intelligent transportation systems:

- **LLM/VLM agents & world models for AV & embodied AI** – trajectory-planning foundation models for behavior simulation (sim agents), world-model construction, and safety-critical scenario generation & closed-loop validation.
- **Robotaxi & AV fleet operations** – routing, dispatching, and repositioning at network scale, with fleets doubling as mobile sensing platforms; via MILP and learning-based combinatorial optimization.
- **Agentic system implementation & deployment** – multi-agent architectures, tool orchestration (MCP), and agent evaluation for production workflows, from smart-city & robotaxi operations to automated research pipelines.
- **Cooperative perception & V2X systems** – from algorithms to real-world field deployment.

EDUCATION

- **Purdue University** Aug.2021 - Dec.2026 (Expected)
Doctor of Philosophy West Lafayette, US
 - Research Area: Connected and Automated Vehicle, Cooperative Perception, Physics-informed Machine Learning, Network Operation and Control
 - GPA: 3.8/4.00
- **Carnegie Mellon University** Aug.2019 - Dec.2020
Master of Science Pittsburgh, US
 - Area: Smart City, Deep Learning, Natural Language Processing
 - Course: Deep Learning, Reinforcement Learning, Geo-Informatics System
 - GPA: 3.87/4.00
- **Tongji University** Aug.2015 - Jun.2019
Bachelor of Science Shanghai, China
 - Major: Civil Engineering (Area of Geo-technical and Underground Engineering)

WORKING EXPERIENCE

- **Inchor(SaferDrive AI)** Dec.2025 - Jun.2026
Simulation Development Engineer Intern Ann Arbor, MI
 - Develop and maintain TeraSim, an open-source city-scale traffic simulation platform for autonomous vehicle safety validation built upon SUMO.
 - Implement a full-stack Driving Foundation Model into product.
 - Lead joint project with NVIDIA, and City of San Jose on Smart City Operation with agent
 - Initial and develop Agent-Sim, an agentic World Model for autonomous driving safety validation.

PATENTS AND PUBLICATIONS

C=CONFERENCE, J=JOURNAL, S=IN SUBMISSION

- [S.1] **Yilin Wang, Yiheng Feng***.(2026) **A Cooperative Perception Based Dynamic Vehicle Routing Framework for Urban Traffic Monitoring**. In submission of *Transportation Research: Part B*.
- [C.1] **Yilin Wang, Yiheng Feng*** (2025). **A Cooperative Perception Based Dynamic Vehicle Routing Framework for Urban Traffic Monitoring**. In *Transportation Research Board Annual Meeting 2026*, Accepted by Poster Presentation.
- [J.1] Hanlin Chen, Vamsi K Bandaru, **Yilin Wang**, Mario A Romero, Andrew Tarko, Yiheng Feng* (2024). **Cooperative Perception System for Aiding Connected and Automated Vehicle Navigation and Improving Safety** in *Transportation Research Record*, 2678(12), 1498-1510. <https://doi.org/10.1177/03611981241252779>
- [J.2] **Y. Wang** and **Y. Feng*** (2024). **IDM-Follower: A Model-Informed Deep Learning Method for Car-Following Trajectory Prediction** in *IEEE Transactions on Intelligent Vehicles*, vol. 9, no. 6, pp. 5014-5020, June 2024, doi: 10.1109/TIV.2024.3367654
- [C.2] **Yilin Wang, Yiheng Feng*** (2024). **Dynamic Routing of Connected and Automated Vehicles for Improving Network Coverage**. In *Transportation Research Board Annual Meeting 2024*, Accepted by Poster Presentation.
- [J.3] Ya-Dong Xue, Wei Zhang, **Yi-Lin Wang***, et al. (2023). **Serviceability evaluation of highway tunnels based on data mining and machine learning: A case study of continental United States** in *Tunneling and Underground Space Technology*, Volume 142, 2023, 105418, ISSN 0886-7798, <https://doi.org/10.1016/j.tust.2023.105418>

PROJECTS

• TeraSim: City-Scale World Model for Safety Testing on AV and Physical AI

Dec.2025 - Present

Core Developer | Internship | SaferDrive AI(Inchor)

- **Overview:** A city-scale simulation platform serving as the foundational infrastructure for agentic world models, enabling naturalistic and adversarial testing of autonomous vehicles at urban scale.
- **Core Module – NNDE (Neural-network Naturalistic Driving Environment):** Behavioral-layer foundation module of TeraSim that integrates a transformer-based motion model to drive city-scale, naturalistic AV behavior generation.
- **Core Module – Agentic Test Engineer:** A multi-agent framework powered by LLMs and VLMs that performs autonomous scenario mutation to surface safety-critical corner cases for AV testing.
- **Milestone:** 1. 2026-03: Active deployment with **NVIDIA** and **City of San Jose** as an *Agentic Incident-to-Operation* pipeline, bringing TeraSim from simulation into real-world city operations. 2. 2026-06: First draft release on Agent-Sim, utilizing LLMs and VLMs for safety critical senario generation.
- **Tech Stack:** Python, SUMO, PyTorch, CUDA, Docker,

• Developing A Cooperative Perception System

Jan.2023 - Present

Project Leader | Funded by Center of Connected and Automated Transportation

- **Phase 1:** A Cooperative Perception System for CAV Navigation and Improving Safety
 - * **Objective:** Implement a cooperative perception system to improve road safety.
 - * **Contribution:**
 1. We build a cooperative perception system connected by V2X system between roadside LiDAR and CAV sensors.
 2. We develop a detecting and navigating algorithm on CAV with by sensor data fusion pipeline.
 3. We build and conduct a field testing demonstrating the safety benefits in detecting occluded VRUs through cooperative perception system.
 - * **Conclusion:** Cooperative perception system helps CAVs to plan safer (i.e., higher post-encroachment time) and smoother (i.e., lower deceleration rates) trajectories.
 - * **Outcome:**
 1. Publication: [Cooperative Perception System for Aiding Connected and Automated Vehicle Navigation and Improving Safety](#)
- **Phase 2:** An Infrastructure-vehicle Cooperative Perception System
 - * **Objective:** Develop a cooperative perception system including multiple vehicles, infrastructure sensors, and drones.
 - * **Contribution:** Extract and processing multi-sensor data (Lidar, Camera, etc.)to vehicle trajectories. Release open-source dataset for futural research.
 - * **Methodology:** Multi-sensor sensor fusion from infrastructure sensor, drone-view camera, and vehicle sensor.
- **Cooperative Perception based Operation for CAVs to Improve Traffic Data Collection** Apr.2023 - Jun.2026
Individual Research Project [🔗]
 - **Project 1:** Dynamic Routing on Connected and Automated Vehicles for Improving Network Coverage
Apr.2023 - Aug.2024
 - * **Research Question:** How to model the CAV network coverage and operate routes to improve data collection.
 - * **Contribution:**
 1. We formulate the CAV routing problem considering the network coverage as one objective.
 2. We propose a heuristic algorithms with greedy search to solve the multi-objective optimization.
 3. We conduct a simulation experiment in a grid network to verify the feasibility and relations between travel cost and network coverage.
 - * **Methodology:** heuristic optimization, SUMO simulation, link-based travel time estimation.
 - * **Outcome:** Accepted by [Transportation Research Board \(TRBAM\) 2024](#) as poster presentation.
 - **Project 2:** A Cooperative Perception based Dynamic Vehicle Routing Framework For Urban Traffic Monitoring
Sept.2024 - Jun.2026
 - * **Research Question:**
 1. A higher granularity of road network representation is needed.
 2. A proper performance metric is needed to evaluate traffic monitoring impact.
 3. There is a dynamic effect between CAV routes decision and partial network observations.
 - * **Contribution:**
 1. We apply Cell Transmission Model (CTM) for traffic state prediction and road network representation considering CAV detection maneuver.
 2. We propose a comprehensive MILP formulation considering the performance of traffic monitoring as well as dynamic vehicle routing in one explict equation.

3. We conduct a microscopic simulation and evaluations under various scenarios.

* **Outcome:** Submit an Under Reviewed paper by Transportation Research: Part B.

• **IDM-Follower: A Physics-Informed Neural Network Model for Trajectory Prediction**

Jan.2022 - Feb.2024

Individual Research Project



◦ **Research Question:** How to integrate explicit car-following model into learning-based model to improve training and performance.

◦ **Contribution:**

1. We introduce a physics-informed neural network (PINN) model that integrates intelligent driving model (IDM).
2. The proposed model exhibits robustness against real-time GPS noise.

◦ **Methodology:** The loss function combined by physical laws and ground-truth difference is utilized to train a customized attention-based VAE model.

◦ **Outcome:**

1. Presentation: Research Presentation on IEEE-IV 2024 at Jeju Island, KR.
2. Publication: **IDM-Follower: A Model-Informed Deep Learning Method for Car-Following Trajectory Prediction**

• **Developing V2X System for Traffic Signal Control**

Sept.2021 - Jun.2026

Multiple Projects, Funded by US-DOT, Michigan DOT, and Leidos Inc.

◦ **Part 1: Smart Intersection Project**

Sept.2021 - Jun.2026

* **Overview:** Implement CV2X system for Transit Signal Priority (TSP) in City of Ann Arbor.

* **Contribution:**

1. Redevelop Multi-Modal Intelligent Traffic Signal System (MMITSS).
2. Maintain and update versions on [road-side](#) and [vehicle-side](#) applications.
3. Field testing, installation, and data processing for buses and real intersections.

◦ **Part 2: Michigan DOT ATCMTD Project**

Feb.2023 - present

* **Overview:** Implementing MMITSS system in [Q-Line](#) in Detroit, MI.

* **Contribution:** Update MMITSS to accommodate NTCIP 1211 operation protocol.

◦ **Part 3: CDA-Sim Update: Integrating MMITSS on a Co-simulation Platform**

Dec.2024 - Jun.2025

* **Overview:** Integrating MMITSS into CARMA-CARLA-SUMO co-simulation.

* **Contribution:**

1. New features on time synchronization, message brokers, and parallel processing to support MMITSS in CDA-Sim platform.
2. Build integration testing and demo with released configuration.

SKILLS

- **Programming Languages:** Python, C++, MATLAB, JavaScript
- **Data Science & Machine Learning:** PyTorch, CUDA, Hugging Face
- **Cloud Technologies:** AWS, SSH
- **DevOps & Version Control:** Linux, Docker, Conda
- **Cooperation Platform & Project Management:** Jira, Slack, GitHub
- **Other Tools & Technologies:** Microsoft Office, Auto-CAD, ArcGIS
- **Research Skills:** Critical Thinking, Research Talk, Comprehensive Writing, Literature Review

HONORS AND AWARDS

• **Honorable Mention**

2018

Mathematical Contest in Modeling of USA



ADDITIONAL INFORMATION

Languages: English (Proficiency), Chinese Mandarin (Native)

Interests: Soccer, Skateboarding, Range and Gaming

REFERENCES

1. **Yiheng Feng**

Assistant Professor, Civil and Construction Engineering
Assistant Director, Center for Road Safety (CRS)
Lyles School of Civil and Construction Engineering, Purdue University
Relationship: Ph.D Advisor

2. **Pingbo Tang**

Associate Professor, Civil and Environment Engineering
Carnegie Mellon University
Relationship: Master Research Advisor

3. **Howie Sun**

Co-Founder, Inchor / SaferDrive AI
Relationship: Supervisor

Contact details for references available upon request.